and learn only a low dimensional weight vector for this user. While this approach has found use in domains multi-task RL with low-rank MDPs Agarwal et al. (2020); Bose et al. (2024b), the main challenge in adapting collaborative ranking to RLHF is the high dimensionality of the item space, which consists of user prompts $x \in \mathcal{X}$ and LLM-generated responses $y \in \mathcal{Y}$, forming items $(x, y) \in \mathcal{X} \times \mathcal{Y}$. User preferences are captured through a limited number of pairwise comparisons $(x, y_c) \succ (x, y_r)$, indicating a preference for y_c over y_r given x. However, modern LLMs typically use vocabulary sizes between 32000-128000 (V) tokens and support context windows ranging from 2048 to 2 million (K) tokens (Touvron et al., 2023; Achiam et al., 2023), the item space scales as $M = V^K$—rendering direct reward basis learning $\mathbf{R} \in \mathbb{R}^{B \times M}$ infeasible.

To address this challenge, we propose a framework that models diverse user preferences through a set of B basis reward functions represented by the Reward Basis $\mathbf{R}_\phi : \mathcal{X} \times \mathcal{Y} \mapsto \mathbb{R}^B$. The individual preference function for user i, $\mathbf{p}_i : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, is defined as:

$$\mathbf{p}_i := \mathbf{w}_i^\top \mathbf{R}_\phi, \tag{7}$$

where $\mathbf{w}_i^\top \in \Delta^{B-1}$ is a normalized weight vector inherent to the user. By leveraging a straightforward low-rank matrix factorization, we efficiently capture diverse user preferences. This approach, overlooked in favor of more complex methods in prior work, highlights the strength of collaborative ranking across diverse applications. The simplicity of our method is a major advantage, enabling significant performance gains (see Sec. 5) while being easy to integrate with various downstream tasks (see Sec. 3.4). The space Φ of the learnable parameters depends on the use case and the sample size available for training. We provide a few examples below.

- **Example 1.** Fine-tuning only the final layer of a pre-trained reward model (Ziegler et al., 2019): Standard transformer-based models output a single scalar reward. To modify the reward model to output a B-dimensional representation, we learn a simple linear transformation on top of the embeddings generated by the pre-final layer, denoted as $\mathbf{e} : \mathcal{X} \times \mathcal{Y} \mapsto \mathbb{R}^D$, while keeping other layers frozen. The reward basis is then defined as:

$$\mathbf{R}_\phi(x, y) = \mathbf{A}\mathbf{e}(x, y), \tag{8}$$

  where $\mathbf{A} \in \mathbb{R}^{B \times D}$ is a learnable matrix that projects $\mathbf{e}(x, y)$ into a B-dimensional space.

- **Example 2.** Similar to Example 1, we modify the pre-trained reward model to output a B-dimensional representation by applying a learnable transformation to the embeddings from the pre-final layer. Instead of using a linear transformation, we train a shallow multi-layer perceptron (MLP) f_ϕ on top of these frozen embeddings (Wang et al., 2024):

$$\mathbf{R}_\phi(x, y) = f_\phi(\mathbf{e}(x, y)), \tag{9}$$

  where ϕ are the parameters of the MLP. This allows for more expressive transformations while keeping the earlier layers of the reward model frozen.

- **Example 3.** Fine-tuning earlier layers with LoRA (Low-Rank Adaptation) (Hu et al., 2021): Instead of keeping the earlier layers frozen, we can allow them to be fine-tuned in a parameter-efficient way using LoRA. As in the previous examples, we first modify the pre-trained reward model to output a B-dimensional representation by applying a learnable transformation to the embeddings from the pre-final layer. However, in addition to learning a transformation on top of these embeddings, we also fine-tune the transformer's earlier layers by introducing low-rank adaptation matrices.

3.3 The LoRe Workflow

In this section, we outline the LoRe workflow in full generality. The specific choices made for our experiments are discussed in Section 5.

Collecting User Preference Data: We assume access to preference feedback data from **seen users**, denoted by the set $\mathcal{U}_{\text{seen}}$. Each user provides a set of labeled pairs of responses, denoted as $\mathcal{D}_i = \{(x, y_c, y_r)\}$, where x is the input, and y_c and y_r are the chosen and rejected options, respectively. The training dataset $\mathcal{D}_{\text{train}}$ is the collection of $\mathcal{D}_i$ of all users in $\mathcal{U}_{\text{seen}}$.

Jointly Learning Basis Reward and User Preference Weights: Assuming that the preference for each user follows the BT Model with their personalized reward function $\mathbf{p}_i$, the learning problem boils down to the following maximum likelihood estimation problem:

$$\min_{\phi:\Phi,\{\mathbf{w}_i\in\Delta^{B-1}\}_{i\in\mathcal{U}_{\text{seen}}}} \sum_{i\in\mathcal{U}_{\text{seen}}} \frac{1}{|\mathcal{D}_i|} \sum_{(x,y_c,y_r)\in\mathcal{D}_i} \ell(\mathbf{w}_i^\top(\mathbf{R}_\phi(x,y_c)-\mathbf{R}_\phi(x,y_r))), \tag{10}$$

where $\ell(\cdot)$ is the logistic loss function as described in Eq. (4).

Once the learner has learned these parameters, they are interested in generalizing to new prompt queries by the users in $\mathcal{U}_{\text{seen}}$, as well as being able to adapt to preferences of new users denoted by the set $\mathcal{U}_{\text{unseen}}$.

Few Shot Learning for New Users (Unseen User Generalization): This type of generalization involves predicting well for users whose preference data was not part of the training data at all, i.e., completely new users with unseen preferences. These users are termed as **unseen users** denoted by the set $\mathcal{U}_{\text{unseen}}$. These users have few interaction data points, that is termed as $\mathcal{D}_{\text{fewshot}}$, and this is used to few-shot learn these users' preferences $\{\mathbf{w}_i\}_{i\in\mathcal{U}_{\text{unseen}}}$ by optimizing Eq. (11).

For any user $i \in \mathcal{U}_{\text{unseen}}$ with few feedback samples denoted by $\mathcal{D}_i \in \mathcal{D}_{\text{fewshot}}$, we estimate their preference $\mathbf{w}_{\text{new}} \in \Delta^{B-1}$, keeping the reward basis $\mathbf{R}_\phi$ fixed from Eq. (10) as follows:

$$\mathbf{w}_{\text{new}} = \underset{\mathbf{w}\in\Delta^{B-1}}{\text{argmin}} \sum_{(x,y_c,y_r)\in\mathcal{D}_i} \ell(\mathbf{w}^\top(\mathbf{R}_\phi(x,y_c)-\mathbf{R}_\phi(x,y_r))). \tag{11}$$

3.4 Personalized Response Generation via Steerable Multi-Objective Alignment

Recently, there has been a growing interest in Multi-Objective Alignment (MOA) in LLMs. Formally, let $\mathbf{R}_\phi : \mathcal{X}\times\mathcal{Y}\to\mathbb{R}^B$ represent B reward functions for different (often conflicting) objectives. For instance, one objective may favor detailed explanations, while another prioritizes conciseness. The goal is to generate responses with varying emphasis, dictated by a weight vector $\mathbf{w}\in\mathbb{R}^B$, yielding the reward function $\mathbf{w}^\top\mathbf{R}_\phi$. A naive approach would be to train a language model for every possible $\mathbf{w}$, but this becomes infeasible as the space of $\mathbf{w}$ is infinite.

Prior work (Wu et al., 2023; Zhou et al., 2023; Rame et al., 2024; Jang et al., 2023; Shi et al., 2024) demonstrates that it is possible to learn only B language models corresponding to these B reward functions and still generate responses for any arbitrary weight $\mathbf{w}$ at inference time. However, *these works assume known reward models* (using off-the-shelf reward models for observable objectives like harmlessness, conciseness, etc.) and *provided preference weights.*

In personalization, objectives are often latent and subjective, requiring learning rather than explicit specification. Furthermore, preferences are inherently subjective and can be difficult to articulate, making it challenging for users to specify precisely what weights they want. Modeling personalized rewards as linear combinations of basis functions enables adaptation to users' implicit preferences. Our research is *orthogonal yet complementary* to this line of work, enabling seamless integration into advances in multi-objective alignment.

4 Related Work and Contributions

Personalized Reward Learning from Human Feedback: This line of research typically captures diversity in user preferences in one of the following ways: 1.**Explicitly categorize users** based on observable traits (e.g., demographics, personality traits) and train separate reward models for each category (Jiang et al., 2023; Zhu et al., 2024; Bose et al., 2024a). This approach is inherently limited in granularity and struggles with cases where user preferences do not align neatly with predefined categories. 2. **Learn per-user reward models** by conditioning on latent representations of user preferences (Poddar et al., 2024; Chen et al., 2024a; Lee et al., 2024). These methods require significant data for each user and do not generalize well to new users with limited feedback. We elaborate on two of these methods next.

Personalized reward modeling for pluralistic alignment (PAL) : (Chen et al., 2024a) assume that the latent preference of each user is modeled by an unknown ideal point $\mathbf{z}_i \in \mathbb{R}^D$. They propose two methods to represent these ideal points:

PAL-A: In this approach, the ideal points factorize as $\mathbf{z}_i = \mathbf{Q}\mathbf{w}_i$, where $\mathbf{Q} \in \mathbb{R}^{D\times B}$ represents B prototype ideal points and $\mathbf{w}_i \in \Delta^{B-1}$ represents weights over these prototypes for user i. Given a pre-trained embedding function $\mathbf{e} : \mathcal{X} \times \mathcal{Y} \mapsto \mathbb{R}^D$, the reward for user i for a prompt response pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$ is given by the Euclidean distance square between $\mathbf{e}(x, y)$ and $\mathbf{Q}\mathbf{w}_i$ in a learnable representation space defined by function $f_\theta : \mathbb{R}^D \mapsto \mathbb{R}^d$ (a shallow MLP with parameters θ):

$$\mathbf{p}_i(x, y) := \|f_\theta(\mathbf{e}(x, y)) - f_\theta(\mathbf{Q}\mathbf{w}_i)\|_2^2. \tag{12}$$

The learnable parameters are $\theta, \mathbf{Q}, \{\mathbf{w}_i\}_{i\in\mathcal{U}_{\text{seen}}}$. This method aims to generalize to unseen users by covering the user space with the prototype matrix.

PAL-B: Here, the user ideal point is a function of the prompt, expressed as $\mathbf{z}_i(x) = \mathbf{G}_\phi(\mathbf{e}(x))\mathbf{w}_i$. The personalized reward function is defined as:

$$\mathbf{p}_i(x, y) := \mathbf{z}_i(x)^\top f_\theta(\mathbf{e}(y)) = \mathbf{w}_i^\top \mathbf{G}_\phi^\top(\mathbf{e}(x)) f_\theta(\mathbf{e}(y)), \tag{13}$$

where $\mathbf{G}_\phi : \mathbb{R}^D \mapsto \mathbb{R}^d \times \mathbb{R}^K$ and $f_\theta : \mathbb{R}^D \mapsto \mathbb{R}^d$. Mathematically, PAL-B is a special case of Eq. (7) where $\mathbf{R}_\phi(x, y) := \mathbf{G}_\psi^\top(\mathbf{e}(x)) f_\theta(\mathbf{e}(y))$. This decomposition of $\mathbf{R}_\phi$ is not novel to PAL-B and has already appeared in a prior work (Wang et al., 2024).

Variational Preference Learning (VPL) (Poddar et al., 2024): The latent preference of each user is denoted by $\mathbf{z}_i \in \mathbb{R}^D$, which is the output of an encoder function $\mathbf{Q}_\theta : \mathcal{D}_i \mapsto \mathbb{R}^D$, that maps the user preference data $\mathcal{D}_i$ to a latent code. Given a pre-trained embedding function $\mathbf{e} : \mathcal{X} \times \mathcal{Y} \mapsto \mathbb{R}^D$, the reward for user i for a prompt response pair $(x, y) \in \mathcal{X} \times \mathcal{Y}$ is given by a learnable function $R_\phi : \mathbb{R}^{2D} \mapsto \mathbb{R}$, as:

$$\mathbf{p}_i(x, y) := \mathbf{R}_\phi(\mathbf{e}(x, y); \mathbf{Q}_\theta(\mathcal{D}_i)). \tag{14}$$

The noise in the latent space encourages the reward model to learn over the entire latent space, which encourages the production of meaningful rewards for unseen users, with the learnable parameters being those of $\mathbf{R}_\phi$ and $\mathbf{Q}_\theta$.

Personalized Response Generation: Following the idea of Direct Preference Optimization (DPO) (Rafailov et al., 2024), which learns a response generation policy (a language model) without explicitly learning a reward function, recent approaches directly model personalized responses. Li et al. (2024) use an encoder to generate latent user embeddings, conditioning a language model on them via DPO. Chen et al. (2024b) personalizes responses through user-specified prompts, while Woźniak et al. (2024) incorporates user information as input features. Other works explore limited personalization settings, such as multiple-choice questions (Zhao et al., 2023), explicit human corrections (Shaikh et al., 2024), and few-shot adaptation with synthetic users (Singh et al., 2025). In Appendix A, we show that the core idea of LoRe naturally extends to response generation without explicit reward learning too.

Evaluating personalized response generation is particularly challenging. Unlike reward learning, where user data can be held out for validation, there is no access to users in offline datasets to label their preferences on newly-generated responses. Instead, evaluations rely on heuristic LLM-based evaluators, which are typically trained using monolithic Bradley-Terry reward models assuming a single global ranking of responses, failing to capture diverse user preferences. Hence, as discussed in Panickssery et al. (2024); Dong et al. (2024), these models reinforce bias; favoring dominant preference patterns while undervaluing minority preferences and struggle with ambiguous or tied preferences, leading to systematic misalignment with real user satisfaction.

Contributions We state our contributions and how we overcome limitations in prior work:

1. **Latent Basis Reward Functions for Personalized Alignment** LoRe introduces a structured approach to personalization by learning a set of primitive reward functions (basis functions) that span the space of individual reward models. Each user's preference is represented as a weighted combination of these basis functions, enabling smooth adaptation without requiring predefined user categories or extensive per-user data.

2. **Decoupled Learning for Efficient and Generalizable Adaptation** LoRe separates the learning of basis reward functions from user-specific weights, enabling rapid few-shot personalization. Once the basis functions are learned, a new user's preferences can be captured with only a small number of interactions, making